CS3 Case Study Rubric

Why am I doing this?

This case study gives you the chance to apply real data science skills to a problem that affects millions of people every day. Energy grid operators have to forecast electricity demand hours in advance to keep the lights on, and weather is one of the most intuitive candidate predictors. You will get hands-on experience working with time-series data and building and comparing forecasting models while practicing making a researched argument about whether a more complex model is actually worth it. The project also builds your ability to communicate data-driven findings clearly.

Learning Objective: Apply time-series modeling and model comparison methods to real-world data to generate and communicate insights.

What am I going to do?

You will use hourly electricity demand data from PJM Interconnection and daily weather observations from NOAA to build two forecasting models. First, you will train a baseline ARIMA model using only past electricity demand values. Then you will build a second model that incorporates weather variables as additional predictors. You will compare the two models using RMSE and MAPE and write a short report interpreting what you found.

Deliverables include:

- A GitHub repository with your scripts, data, and outputs
- At least two forecasting models compared on RMSE and MAPE
- A 1-2 page written report (PDF) summarizing your findings

Tips for Success:

- Read the repository README carefully before starting. It walks you through the full pipeline step by step.
- Spend real time on creating exploratory visualizations before modeling. Understanding seasonal and weekly patterns in the data will make your modeling decisions much more intuitive.
- Don't just report RMSE and MAPE. Make sure to explain what the difference between your models actually tells you.
- Weather may not be the primary driver of electricity demand. That is a valid and interesting finding if you can explain why.

Formatting	One GitHub repository submitted via canvas link containing: README.md, LICENSE.md (MIT), SCRIPTS folder, DATA folder, OUTPUT folder. A 1-2 page written report (PDF) following the section structure below:

<table>
<tr><td></td><td>Executive Summary
Data Acquisition & Description
Methodology
Results & Visualizations
Interpretation & Discussion
Reproducibility & Repository Quality
Writing should be clear and accessible to a general audience. All figures must be labeled and readable.</td></tr>
<tr><td>README.md</td><td><u>Goal:</u> This file serves as an orientation to everyone who comes to your repository. Use markdown headers to divide content.
Section 1: Explain the details of the project, what you are doing, and why it is important
Section 2: Provide an outline or tree illustrating the hierarchy of folders and subfolders in your repository, listing the files stored in each
Section 3: Summarize your findings, explain what outputs you generated, what we can learn from them, and how those findings could be used</td></tr>
<tr><td>LICENSE.md</td><td><u>Goal:</u> This file explains to a visitor the terms under which they may use and cite your repository. A license is already provided in the original repository. If it is deleted, use the MIT license.</td></tr>
<tr><td>SCRIPTS folder</td><td><u>Goal:</u> This folder contains all the source code for your project. Place all code scripts used for this project into this folder. These should include the initial case study scripts and any additional code you generate. All script files should include header comments at the beginning explaining what the script does and what anyone running it should know. Throughout your scripts, include comments explaining what each command or sequence of commands accomplishes and why.</td></tr>
<tr><td>DATA folder</td><td><u>Goal:</u> This folder contains all of the data for this project. You should include both the raw source files (PJM load data and NOAA weather data) and the cleaned, merged dataset you did the analysis with. If the data files are too large for GitHub, include a single file that explains how to obtain the dataset and describes what preprocessing steps are required to reproduce the final merged file.</td></tr>
<tr><td>Output folder</td><td><u>Goal:</u> This folder contains all of the output generated by your project, including figures and tables. This folder should contain your exploratory visualizations (time-series plots of demand and weather, and correlation heatmap) and your forecast comparison plots for both models. It should also include the reflection where you summarize what you learned and took away from this project, including one modeling decision you made and why, and one limitation of your analysis, formatted as a PDF, one-page maximum.</td></tr>
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